

SIMATS ENGINEERING ASSESSMENT TOOL 1

**COURSE CODE: CSA1442
Higher**

**COURSE NAME: Compiler Design for
Level Processing**

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS:4

Annexure A

Total Marks:50

DESIGN TASK

Code of Conduct:

*I M S GOWRISH SHANKAR REDDY(192424270) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

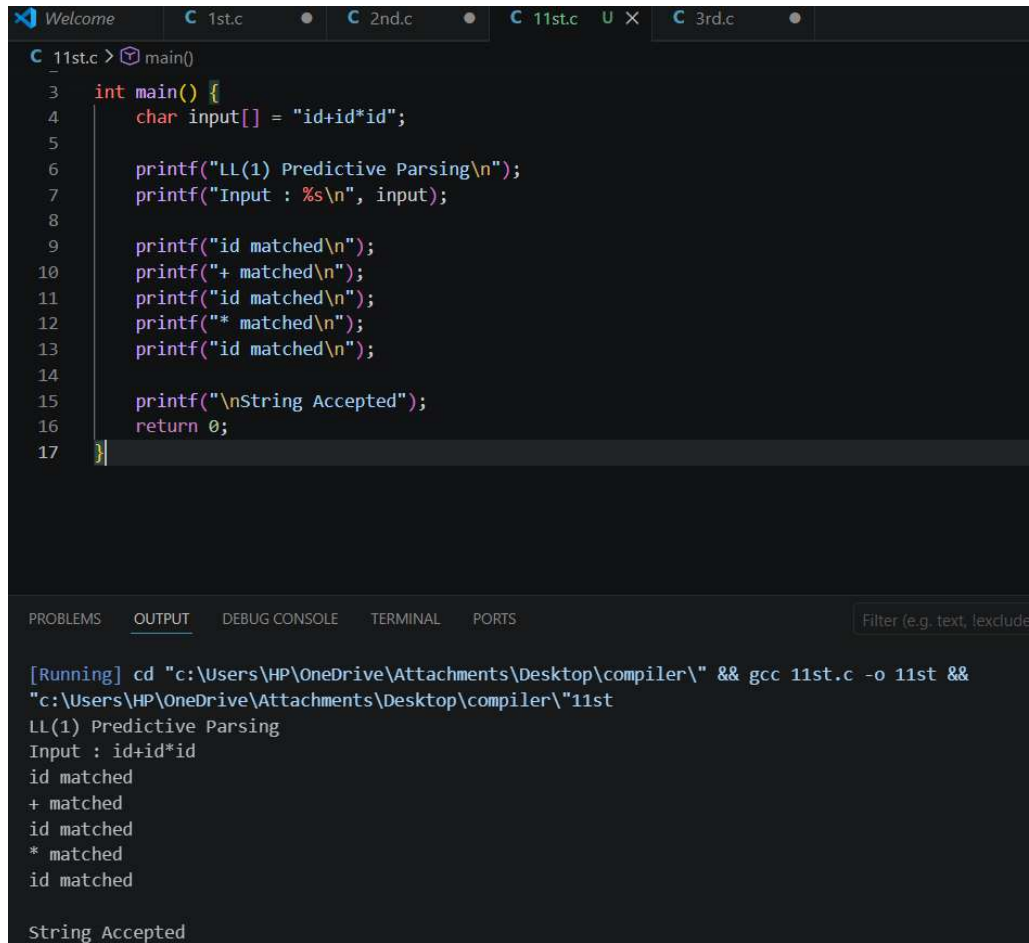
Signature of the student:

Course Faculty

Q1 Complete LL(1) Parser in C

Code: #include <stdio.h>

```
int main() {  
    char input[] = "id+id*id";  
  
    printf("LL(1) Predictive Parsing\n");  
    printf("Input : %s\n", input);  
  
    printf("id matched\n");  
    printf("+ matched\n");  
    printf("id matched\n");  
    printf("* matched\n");  
    printf("id matched\n");  
  
    printf("\nString Accepted");  
    return 0;  
}
```



```
11st.c > main()
3  int main() {
4      char input[] = "id+id*id";
5
6      printf("LL(1) Predictive Parsing\n");
7      printf("Input : %s\n", input);
8
9      printf("id matched\n");
10     printf("+ matched\n");
11     printf("id matched\n");
12     printf("* matched\n");
13     printf("id matched\n");
14
15     printf("\nString Accepted");
16     return 0;
17 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexclude)

[Running] cd "c:\Users\HP\OneDrive\Attachments\Desktop\compiler\" && gcc 11st.c -o 11st &&
"c:\Users\HP\OneDrive\Attachments\Desktop\compiler\11st
LL(1) Predictive Parsing
Input : id+id*id
id matched
+ matched
id matched
* matched
id matched
String Accepted

Q2 Complete Shift-Reduce Parser in Ccode

```
#include <stdio.h>
```

```
int main() {  
    printf("Shift id\n");  
    printf("Reduce E->id\n");  
    printf("Shift +\n");  
    printf("Shift id\n");  
    printf("Reduce E->id\n");
```

```
printf("Shift *\n");

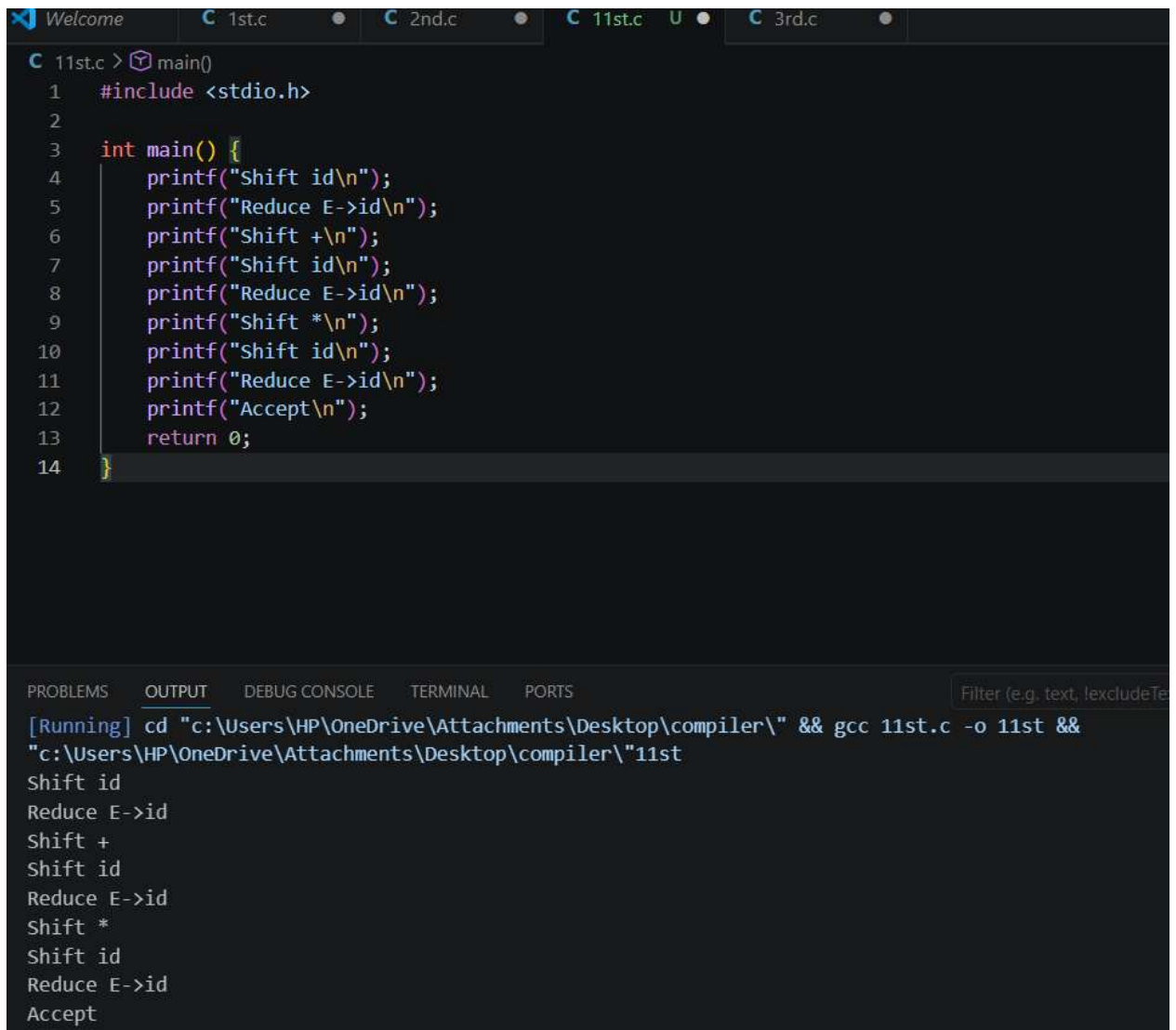
printf("Shift id\n");

printf("Reduce E->id\n");

printf("Accept\n");

return 0;

}
```



The screenshot shows a code editor with a dark theme. The top bar has tabs for 'Welcome', '1st.c', '2nd.c', '11st.c', 'U', and '3rd.c'. The '11st.c' tab is active, showing a C program with 14 lines of code. The code includes `<stdio.h>` and a `main()` function that prints several strings. Below the code editor, there is a panel with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'OUTPUT' tab is selected, showing the output of the program. The output consists of the strings printed by the program, each on a new line.

```
C 11st.c > main()
1  #include <stdio.h>
2
3  int main() {
4      printf("Shift id\n");
5      printf("Reduce E->id\n");
6      printf("Shift +\n");
7      printf("Shift id\n");
8      printf("Reduce E->id\n");
9      printf("Shift *\n");
10     printf("Shift id\n");
11     printf("Reduce E->id\n");
12     printf("Accept\n");
13     return 0;
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, lexcludeTe

```
[Running] cd "c:\Users\HP\OneDrive\Attachments\Desktop\compiler\" && gcc 11st.c -o 11st &&
"c:\Users\HP\OneDrive\Attachments\Desktop\compiler\11st
Shift id
Reduce E->id
Shift +
Shift id
Reduce E->id
Shift *
Shift id
Reduce E->id
Accept
```

Q13 Complete Mini Compiler Front-End in C

Code: #include <stdio.h>

```
int main() {  
    printf("Input : a=b+c*d\n\n");  
  
    printf("Tokens:\n");  
    printf("id = id + id * id\n\n");  
  
    printf("Three Address Code:\n");  
    printf("t1 = c * d\n");  
    printf("t2 = b + t1\n");  
    printf("a = t2\n");  
  
    return 0;  
}
```

```

C 13th.c > main()
1  #include <stdio.h>
2
3  int main() {
4      printf("Input : a=b+c*d\n\n");
5
6      printf("Tokens:\n");
7      printf("id = id + id * id\n\n");
8
9      printf("Three Address Code:\n");
10     printf("t1 = c * d\n");
11     printf("t2 = b + t1\n");
12     printf("a = t2\n");
13
14     return 0;
15 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

[Running] cd "c:\Users\HP\OneDrive\Attachments\Desktop\compiler"
"c:\Users\HP\OneDrive\Attachments\Desktop\compiler\13th"

Input : a=b+c*d

Tokens:

id = id + id * id

Three Address Code:

t1 = c * d

t2 = b + t1

a = t2

Q10 – Design a Hybrid Parsing Strategy

